

REQUIREMENT ANALYSIS

Objective

This Customer Journey Map explains how users request access, how approvals are processed, and how roles and permissions are assigned securely through automated workflows. It improves security, efficiency, and user experience.

3.1 Customer Journey Map

Stage | User Activity | System Response | User Experience

Login | User logs into the system | Authentication is verified | Secure and quick login

Access Request | User submits a request for a role or group | Request is created and forwarded to the approver | Easy request submission

Approval | Manager/Admin reviews the request | Workflow processes the approval | Faster approval process

Role & Group Assignment | Approved role and group are assigned | System updates user permissions | Secure access assignment

Access Verification | User accesses authorized modules | ACL validates user permissions | Controlled and secure access

Benefits

Reduces manual effort.

Improves security and compliance.

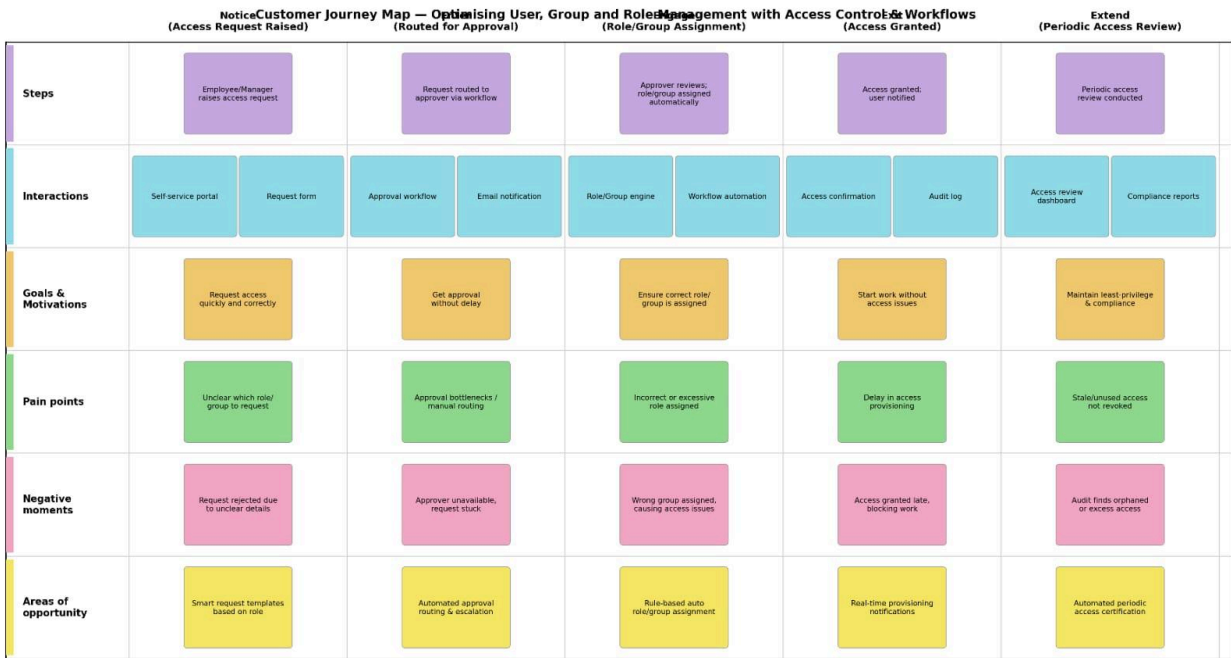
Speeds up approval workflow.

Ensures accurate role assignment.

Enhances user satisfaction.

Conclusion

The Customer Journey Map demonstrates how access requests are processed efficiently using role management, access control, and automated workflows, ensuring secure and reliable user access.



3.2Solution requirement :

Team ID	SWTID-2026-5870
Project Name	Optimizing User, Group, and Role Management with Access Control and Workflows

Functional requirements:

Following are the functional requirements :

FR.No	Functional requirements	Sub requirements
FR-1	Requirement Analysis & Planning	Business Objectives Functional Scope Stakeholder Mapping Execution Roadmap
FR-2	Core Platform Setup (Users, Groups, Roles)	Creation of Users, Creation of Groups, Creation of Roles
FR-3	Data & Structural Configuration	Create Project Table, Create Task Table 2
FR-4	Access Assignment & Governance Enablement	Assign Roles to Alice (Project Manager), Assign Roles to Bob (Team Member)
FR-5	Application Security & Access Enforcement	Assign Table Access to Application, Create ACLs for Task Table 2
FR-6	Automation & Workflow Optimization	Create a Flow to Assign, Operations Ticket to Group
FR-7	Testing & Validation	Workflow Execution Testing
FR-8	Project Conclusion	Conclusion

Non-functional requirements:

Following are the non-functional requirements

FR No	Non-Functional Requirement	Description
NFR-1	Usability	The system provides a simple, intuitive, and user-friendly interface that enables administrators and users to efficiently manage users, groups, roles, permissions, and workflows with minimal training.
NFR-2	Security	The system ensures secure authentication, role-based access control (RBAC), authorization, and encryption to protect user data and prevent unauthorized access to resources.
NFR-3	Reliability	The system performs consistently without failures, ensuring accurate user-role assignments, workflow execution, and uninterrupted access to authorized resources.
NFR-4	Performance	The system provides fast response times for user authentication, role assignment, permission validation, and workflow processing, even when multiple users access the system simultaneously.
NFR-5	Availability	The system is available 24/7 with minimal downtime, ensuring users can securely access the application and perform role-based operations whenever required.
NFR-6	Scalability	The system efficiently supports an increasing number of users, groups, roles, and workflows without affecting performance, making it suitable for growing organizations.

3.3Data flow diagram:

Data Flow Diagram (DFD) for the Optimizing User, Group, and Role Management with Access Control and Workflows project illustrates how data moves through the system. The administrator creates and manages users, groups, and roles, which are stored in the database. Access control validates user permissions before allowing access to resources. Workflow processes handle user requests such as role assignment, approval, and group updates. Finally, the system updates the database and provides the appropriate access based on the assigned roles and permissions.

User Stories

User Story 1

As an Employee, I want to request system access so that I can perform my assigned tasks.

User Story 2

As an IT Administrator, I want to create and manage users, groups, and roles so that access permissions are assigned correctly.

User Story 3

As a Manager, I want to approve or reject access requests so that only authorized users receive access.

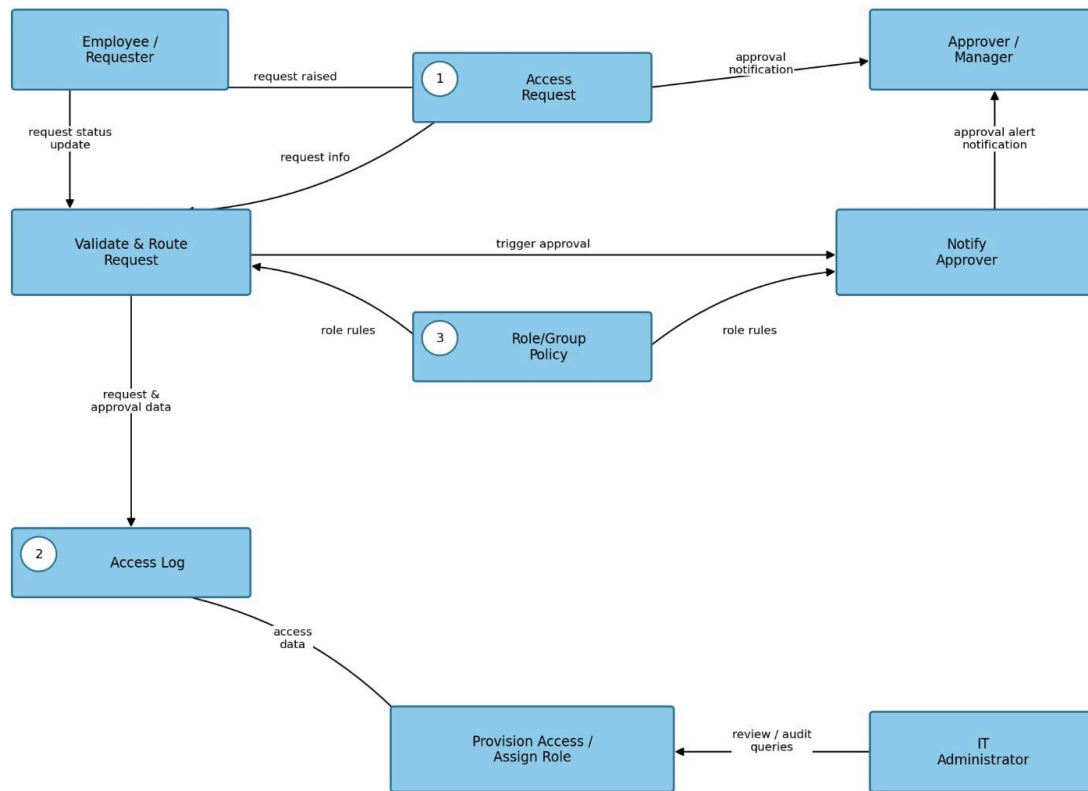
User Story 4

As a System Administrator, I want to implement Role-Based Access Control (RBAC) and ACLs so that unauthorized access is prevented.

User Story 5

As an Auditor, I want to view access logs and reports so that security and compliance can be monitored.

Example: DFD Level 0 — Optimising User, Group & Role Management with Access Control & Workflows



3.4 Technology Stack

This project is developed on the ServiceNow platform using JavaScript for scripting and MySQL as the backend database. It uses Flow Designer/Workflow Editor for automation, Access Control Lists (ACLs) and Role-Based Access Control (RBAC) for security, and Application Studio for development and customization.

Front-End

ServiceNow Workspace

Service Portal

UI Builder

Catalog Forms

Back-End

ServiceNow Platform

Flow Designer

Business Rules

Script Includes

Glide APIs

Database

sys_user (Users)

sys_user_group (Groups)

sys_user_has_role (Role Mapping)

sys_user_grmember (Group Members)

sys_user_role (Roles)

Security

Access Control Lists (ACLs)

Role-Based Access Control (RBAC)

User Criteria

Data Security Policies

Workflow & Automation

Flow Designer

Approval Policies

Email Notifications

Event Management

Development Tools

ServiceNow Studio

App Engine Studio

Personal Developer Instance (PDI)

Update Sets

Conclusion

The proposed solution uses the ServiceNow platform to optimize user, group, and role management through secure access control, automated workflows, and centralized administration. This improves operational efficiency, reduces manual effort, strengthens security, and ensures better compliance.

